$$\underbrace{r_1(t)}_{\text{ESN}_1\text{ state}} \xrightarrow{\text{readout}} y_1(t) \qquad \underbrace{r_2(t)}_{\text{ESN}_2\text{ state}} \xrightarrow{\text{readout}} y_2(t)$$

$$\downarrow \qquad \text{couple} \qquad \downarrow$$

$$u_1 = y_1 + \alpha(y_2 - y_1) \qquad u_2 = y_2 + \alpha(y_1 - y_2)$$

$$\downarrow \qquad \text{reservoir update} \qquad \downarrow$$

$$r_1(t+1) \qquad r_2(t+1)$$

ESN1 : $y_1(t) = W_{out}^T r_1(t) + b_{out}$

ESN2 : $y_2(t) = W_{out}^T r_2(t) + b_{out}$

coupling :

$$u_1(t) = y_1(t) + \alpha\left[y_2(t) - y_1(t)\right]$$
$$u_2(t) = y_2(t) + \alpha\left[y_1(t) - y_2(t)\right]$$

ESN update :

$$r_1(t+1) = (1-\lambda)\, r_1(t) + \lambda \tanh\left[W_{res}\, r_1(t) + W_{in}\, u_1(t) + b_{res}\right]$$
$$r_2(t+1) = (1-\lambda)\, r_2(t) + \lambda \tanh\left[W_{res}\, r_2(t) + W_{in}\, u_2(t) + b_{res}\right]$$